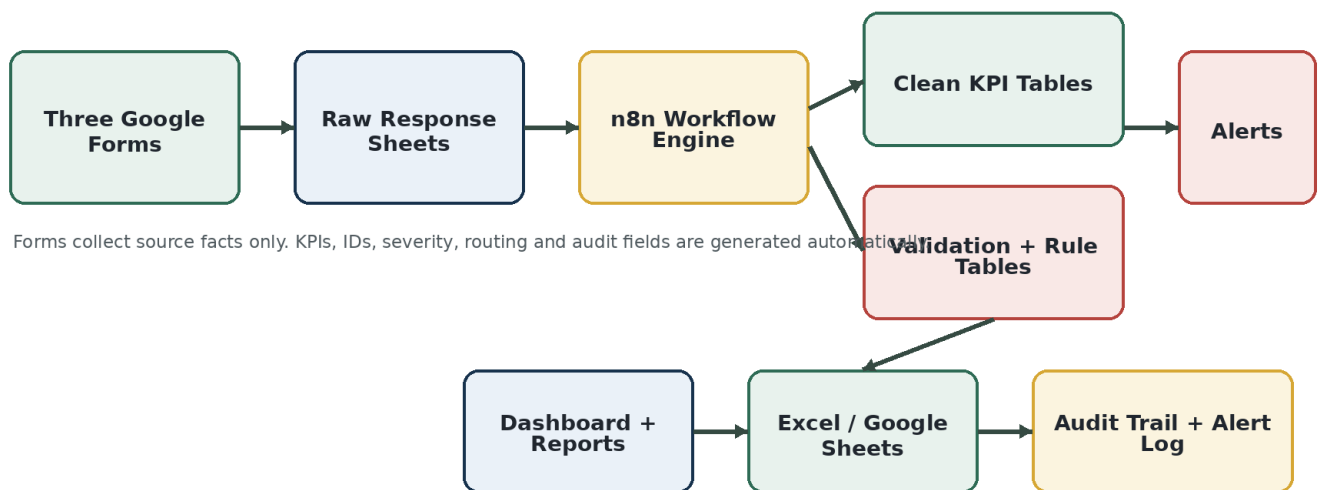


DCP KPI DATA COLLECTION AND n8n AUTOMATION BLUEPRINT

Three Google Forms, Data Mapping, Validation Logic, Exception Routing and Reporting Workflows

DCP KPI Data Collection and Automation Architecture



Proof-of-concept design for Dangote Cement Plc (DCP) using simulated operational data

1. Purpose and Recommended Operating Model

This document specifies the three Google Forms required to collect daily production, downtime and exception-resolution data, and explains how n8n should transform those submissions into clean KPI tables, validation records, stakeholder alerts and management reports.

Recommended model

Use Google Forms only for source facts entered by operators. Do not ask users to calculate Production Achievement, Production Variance, Availability, Severity or Exception Flags. n8n should generate those fields consistently after every submission.

Recommended live data flow

- Each Google Form writes responses to its own raw response tab in one Google Sheets workbook.
- n8n watches each raw sheet for newly added rows, standardizes field names, calculates KPIs and appends clean records to the appropriate operational table.
- Validation rules create Data_Quality_Validation records and exception alerts.
- Critical alerts are routed immediately to the responsible team and escalated to the Plant Manager.
- Daily, weekly and monthly reports are generated from the clean tables, not directly from the raw Form responses.

Recommended workbook tabs

Tab	Purpose
Plant_Master	Static plant and line reference data.
Daily_Production	Clean daily production records and calculated production KPIs.
Downtime	Clean incident records and calculated runtime/availability KPIs.
Data_Quality_Validation	Automated validation results, issue types, scores and exception flags.
Qualitative_Exception_Logic	Rules connecting KPI exceptions to likely causes, teams and actions.
Alert_Log	Every alert sent, recipient, status, acknowledgement and closure details.
Form_Daily_Production_Raw	Raw responses from Form One.
Form_Downtime_Raw	Raw responses from Form Two.
Form_Exception_Closure_Raw	Raw responses from Form Three.

2. Google Form One - Daily Production Submission

Form title

DCP Daily Production Submission

Purpose: one consolidated record per plant and production line for each production day. The form captures source production values and contextual facts; n8n calculates the KPIs.

Form description

Submit one consolidated production record for each plant and production line at the end of every operating day. Enter only source values. KPI calculations and validation checks will be completed automatically.

Paramount questions

Question	Type	Required	Options / Validation
Reporting Date	Date	Yes	Production date.
Plant ID	Dropdown	Yes	OBJ, IBS, GBK.
Production Line	Dropdown	Yes	Line 1, Line 2, Line 3, Line 4.
Reporting Period	Dropdown	Yes	Daily Total.
Submitted By	Short answer	Yes	Full name.
Staff Role	Dropdown	Yes	Production Officer; Shift Supervisor; Reporting Officer; Process Engineer; Other.
Shift	Dropdown	Yes	Day; Night; Consolidated Daily Total.
Clinker Target (tons)	Number	Yes	Zero or greater.
Clinker Actual Output (tons)	Number	Yes	Zero or greater.
Cement Target (tons)	Number	Yes	Greater than zero.
Cement Actual Output (tons)	Number	Yes	Zero or greater.
Packing Actual Output (tons)	Number	Yes	Zero or greater.
Was production interrupted?	Multiple choice	Yes	Yes; No.
Was any production figure estimated?	Multiple choice	Yes	Yes; No.
Submission Status	Multiple choice	Yes	Final; Provisional.
Known Production Constraint	Dropdown	Yes	None; Mechanical Failure; Electrical Fault; Power Interruption; Process Instability; Raw Material Shortage; Bag Shortage; Quality Hold; Packing Delay; Other.
Additional Operational Comment	Paragraph	No	Brief explanation.
Supervisor / Reviewer Email	Short answer	Yes	Email validation.

Fields generated automatically by n8n

Generated field	Logic / purpose
Daily_Production_ID	Unique record key, for example DP-20260714-OBJ-L2.

Generated field	Logic / purpose
Production_Achievement_%	Cement Actual Output / Cement Target x 100.
Production_Variance_Tons	Cement Actual Output - Cement Target.
Clinker_Achievement_%	Clinker Actual Output / Clinker Target x 100.
Packing_Gap_Tons	Packing Actual Output - Cement Actual Output.
Validation_Result	Passed, Warning or Failed.
Data_Quality_Score_%	Score based on completeness, validity and consistency checks.
Exception_Flag	Yes when any configured threshold or quality rule fails.
Severity	Low, Medium, High, Critical or Very Critical.

3. Google Form Two - Downtime Incident Report

Form title

DCP Downtime Incident Report

Purpose: capture an interruption as soon as it occurs, preserve the incident context and provide the raw timestamps required to calculate downtime and availability.

Form description

Submit this form immediately when equipment or a production process stops, becomes unavailable, or operates below its expected capacity. The system will calculate downtime, availability, severity and alert routing.

Section A - Incident identification

Question	Type	Required	Options / Validation
Incident Date	Date	Yes	Date downtime began.
Downtime Start Time	Time	Yes	Actual stoppage time.
Plant ID	Dropdown	Yes	OBJ, IBS, GBK.
Production Line	Dropdown	Yes	Line 1, Line 2, Line 3, Line 4.
Process Stage	Dropdown	Yes	Mining; Crushing; Raw Milling; Preheating/Pre-calcination; Kiln/Clinker; Clinker Cooling; Cement Grinding; Silo; Packing/Dispatch; Utilities.
Equipment Name	Dropdown	Yes	Use standardized equipment names.
Equipment ID	Short answer	Yes	Example: OBJ-L2-KLN.
Reported By	Short answer	Yes	Full name.
Staff Role	Dropdown	Yes	Operator; Shift Supervisor; Production Officer; Maintenance Technician; Electrical Technician; Process Engineer; Other.

Section B - Downtime details

Question	Type	Required	Options / Validation
Downtime Status	Multiple choice	Yes	Ongoing; Equipment Restored.
Downtime End Date	Date	Conditional	Required when Equipment Restored is selected.
Downtime End Time	Time	Conditional	Required when Equipment Restored is selected.
Planned Runtime for the Day	Number	Yes	Between zero and twenty-four hours.
Downtime Type	Dropdown	Yes	Planned; Unplanned.
Downtime Category	Dropdown	Yes	Mechanical; Electrical; Process; Utilities/Power; Warehouse/Materials; Packing/Logistics; Quality; Safety; Other.
Downtime Reason	Dropdown	Yes	Bearing Failure; Belt Failure; Motor Overload; Power Trip; Generator Delay; Instrument Failure; Blockage;

Question	Type	Required	Options / Validation
			Kiln Instability; Feed Interruption; Bag Shortage; Raw Material Shortage; Planned Maintenance; Quality Hold; Packer Jam; Conveyor Failure; Other.
Responsible Area	Dropdown	Yes	Mechanical Maintenance; Electrical; Production; Process Engineering; Utilities; Warehouse; Packing; Logistics; Quality Control; IT/Data Reporting; Other.
Is production affected?	Multiple choice	Yes	Yes; No.
Is the entire line affected?	Multiple choice	Yes	Yes; Partially; No.
Estimated Output Loss (tons)	Number	No	Zero or greater.
Immediate Action Taken	Paragraph	Yes	Brief operational response.
Immediate Safety Risk?	Multiple choice	Yes	Yes; No.
Initial Priority	Dropdown	Yes	Low; Medium; High; Critical.
Supervisor Email	Short answer	Yes	Email validation.

Fields generated automatically by n8n

Generated field	Logic / purpose
Downtime_ID	Unique incident key.
Daily_Production_ID	Links incident to the relevant daily production record.
Downtime_Hours	End timestamp - start timestamp. Ongoing incidents remain provisional.
Actual_Runtime_Hours	Planned Runtime - Downtime Hours.
Availability_%	Actual Runtime / Planned Runtime x 100.
Calculated_Severity	Determined from duration, production impact and safety risk.
Exception_Flag	Yes when availability, downtime or safety thresholds fail.
Alert_ID	Unique alert key for follow-up and closure.
Alert_Status	Open, Acknowledged, In Progress, Monitoring, Resolved or Escalated.
Notification_Timestamp	Time the responsible stakeholder was notified.

4. Google Form Three - Exception Update and Closure

Form title

DCP Exception Update and Closure Form

Purpose: acknowledge alerts, document confirmed causes and corrective action, update unresolved incidents, close resolved exceptions and preserve an auditable response trail.

Form description

Use this form to acknowledge, update, escalate or close a KPI, downtime or data-quality exception. Enter the Alert ID included in the notification. A prefilled link should be sent by n8n whenever possible.

Section A - Exception identification

Question	Type	Required	Options / Validation
Alert ID	Short answer	Yes	Provided in the notification.
Downtime ID	Short answer	Conditional	Required for downtime exceptions.
Daily Production ID	Short answer	Conditional	Required for production exceptions.
Plant ID	Dropdown	Yes	OBJ, IBS, GBK.
Production Line	Dropdown	Yes	Line 1, Line 2, Line 3, Line 4.
Exception Type	Dropdown	Yes	Production; Downtime; Availability; Data Quality; Packing; Other.
Responsible Team	Dropdown	Yes	Plant Management; Production; Mechanical; Electrical; Process; Utilities; Warehouse; Packing; Logistics; Quality; IT/Data Reporting.
Person Responding	Short answer	Yes	Full name.
Respondent Role	Short answer	Yes	Job role.

Section B - Response status

Question	Type	Required	Options / Validation
Update Type	Multiple choice	Yes	Acknowledgement; Progress Update; Resolution and Closure; Escalation.
Current Status	Dropdown	Yes	Open; Acknowledged; In Progress; Monitoring; Resolved; Escalated.
Response Date	Date	Yes	Current response date.
Response Time	Time	Yes	Current response time.
Has the issue been investigated?	Multiple choice	Yes	Yes; No; Investigation Ongoing.
Confirmed Root Cause Category	Dropdown	Yes	Mechanical; Electrical; Process; Utilities/Power; Materials/Warehouse; Packing/Logistics; Data Entry; Sensor/System; Quality; Human Error; Not Confirmed; Other.
Confirmed Root Cause	Paragraph	Yes	Clear explanation.

Question	Type	Required	Options / Validation
Corrective Action Taken	Paragraph	Yes	Work completed so far.
Further Action Required?	Multiple choice	Yes	Yes; No.
Follow-up Owner	Short answer	Conditional	Required when further action is needed.
Follow-up Deadline	Date	Conditional	Required when further action is needed.

Section C - Resolution details

Question	Type	Required	Options / Validation
Equipment / Process Restored?	Multiple choice	Yes	Yes; No; Not Applicable.
Restoration Date	Date	Conditional	Required when restored.
Restoration Time	Time	Conditional	Required when restored.
Actual Output Loss (tons)	Number	No	Zero or greater.
Corrected KPI Value	Number	Conditional	Required for data-quality correction.
Corrected Unit	Dropdown	Conditional	Metric Tons; Hours; Percentage; kWh; Other.
Was the original record corrected?	Multiple choice	Yes	Yes; No; Not Applicable.
Is management approval required?	Multiple choice	Yes	Yes; No.
Approved / Reviewed By	Short answer	Conditional	Supervisor or manager.
Closure Recommendation	Dropdown	Yes	Close Alert; Continue Monitoring; Escalate; Reopen Investigation.
Final Closure Comment	Paragraph	Yes	Brief final update.
Supporting File	File upload	No	Photo, work order, screenshot or inspection record.

Fields updated automatically by n8n

Updated field	Logic / purpose
Resolution_Status	Current lifecycle state of the exception.
Acknowledgement_Timestamp	First response time from responsible team.
Resolution_Timestamp	Time the issue is confirmed as resolved.
Final_Downtime_Hours	Recalculated from confirmed restoration time.
Final_Availability_%	Recalculated using final downtime.
Confirmed_Responsible_Area	Final ownership after investigation.
Alert_Closure_Status	Closed, Monitoring, Escalated or Reopened.
Response_Time_Minutes	Elapsed time between alert and acknowledgement.

5. How to Configure the Google Forms and Response Sheets

Step-by-step setup

- Create the three Forms using the exact titles and field specifications in this document.
- Create one Google Sheets workbook named DCP_KPI_Operational_Input.
- For each Form, open Responses > Select destination for responses > Select existing spreadsheet, and choose the same workbook. Google Forms will create a separate response sheet for each Form. [1]
- Rename the response tabs to Form_Daily_Production_Raw, Form_Downtime_Raw and Form_Exception_Closure_Raw.
- Do not add manual calculations inside the raw response tabs. Preserve them as an immutable intake layer.
- Create the clean operational tabs and reference tabs listed earlier.
- Protect the formula and reference tabs so operators only submit through Forms.

Data governance rule

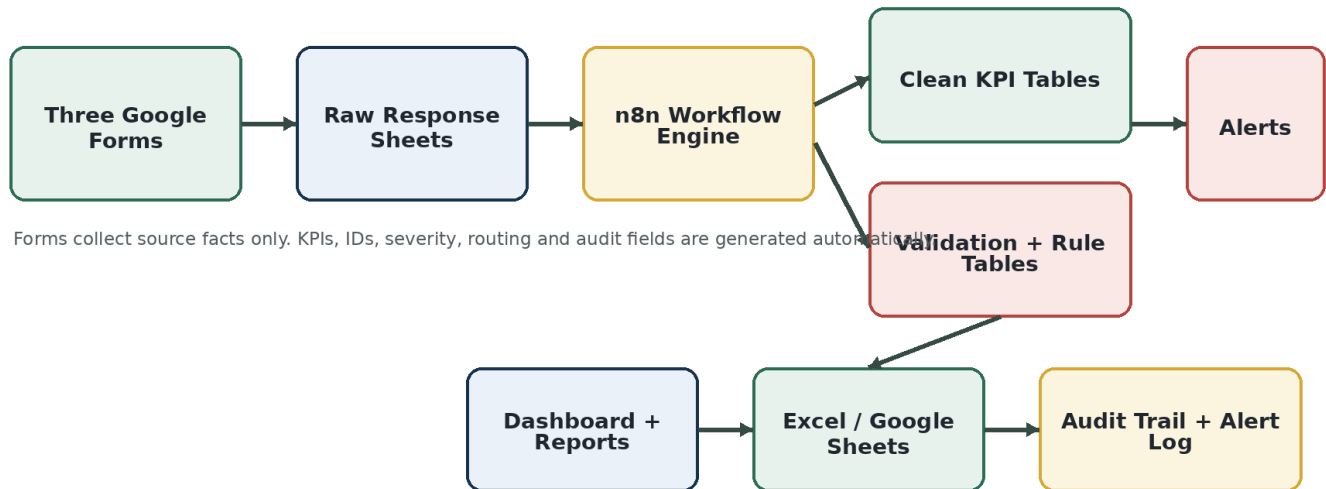
Raw response tabs preserve the original submission. Corrections should be made through the Exception Update and Closure Form, allowing n8n to update the clean table while retaining the original evidence and audit history.

Google Form settings

- Collect respondent email addresses and automatically retain the Form submission timestamp.
- Do not limit users to one response because each plant, line and incident requires repeated submissions.
- Use dropdowns or multiple-choice fields for plant IDs, lines, categories, equipment and responsible areas.
- Apply numeric validation to output, runtime and loss fields.
- Do not allow operators to enter calculated KPI values.
- Use conditional sections in the Downtime and Closure Forms for restoration and follow-up fields.
- Restrict file upload access according to the organization account and security policy.

6. How n8n Connects to the Forms and Data Tables

DCP KPI Data Collection and Automation Architecture



Connection principle

Google Forms does not need to connect directly to n8n. Each Form writes a row to its linked Google Sheets response tab. The n8n Google Sheets Trigger can start a workflow when a row is added, updated, or added/updated. The Google Sheets action node can then read, append or update rows in the clean tables. [2][3]

Credentials and access

- Create or sign in to the n8n instance.
- In n8n, create Google Sheets credentials using the supported Google OAuth2 authentication flow.
- Authenticate with the Google account that owns or can edit the DCP_KPI_Operational_Input workbook.
- In each Google Sheets Trigger node, select the workbook, the relevant raw response sheet and the Row Added event.
- Test the trigger by submitting one sample Form response, then confirm that n8n receives the field names and values.
- Activate the workflow only after field mapping, validation and exception routing tests pass.

Recommended production data store

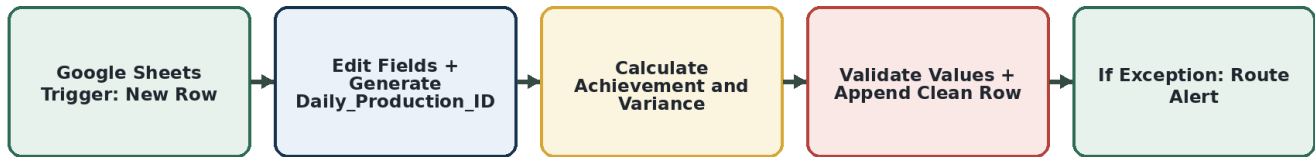
Fastest prototype

Keep Google Sheets as the live data store during the competition. Export the complete workbook to Excel when required for submission, Power BI import or archival.

Where a live Excel workbook is mandatory, store the workbook in OneDrive or SharePoint and use n8n's Microsoft Excel node to add and retrieve table rows. [7] A local desktop Excel file is not a reliable cloud automation source unless n8n is self-hosted with controlled file access.

7. n8n Workflow A - Daily Production Submission

Workflow A - Daily Production Submission



Node-by-node design

Step	n8n node	Configuration / output
1	Google Sheets Trigger	Watch Form_Daily_Production_Raw for Row Added.
2	Edit Fields / Set	Rename Google Form question headings to the clean column names used in Daily_Production.
3	Code or expressions	Generate Daily_Production_ID and calculate Production Achievement, Production Variance, Clinker Achievement and Packing Gap.
4	If nodes	Check required fields, valid plant/line combinations, non-negative values and target values greater than zero. The If node splits the workflow into valid and invalid branches. [4]
5	Google Sheets - Append Row	Append the standardized record to Daily_Production.
6	Google Sheets - Append Row	Append validation outcome to Data_Quality_Validation.
7	If - Exception Flag	Check whether Production Achievement, Production Variance or validation rules breach the proposed thresholds.
8	Switch	Route the exception to Mechanical, Electrical, Production, Warehouse, Data Reporting or another stakeholder based on Responsible_Area. The Switch node supports multiple conditional outputs. [5]
9	Email / Microsoft Teams	Send the stakeholder highlight report.
10	Google Sheets - Append Row	Write Alert_ID, recipients, severity and sent timestamp to Alert_Log.

Production calculations

KPI	Calculation
Production Achievement (%)	$\text{Cement_Actual_Tons} / \text{Cement_Target_Tons} \times 100$
Production Variance (tons)	$\text{Cement_Actual_Tons} - \text{Cement_Target_Tons}$
Clinker Achievement (%)	$\text{Clinker_Actual_Tons} / \text{Clinker_Target_Tons} \times 100$
Packing Gap (tons)	$\text{Packing_Actual_Tons} - \text{Cement_Actual_Tons}$

Illustrative exception rules

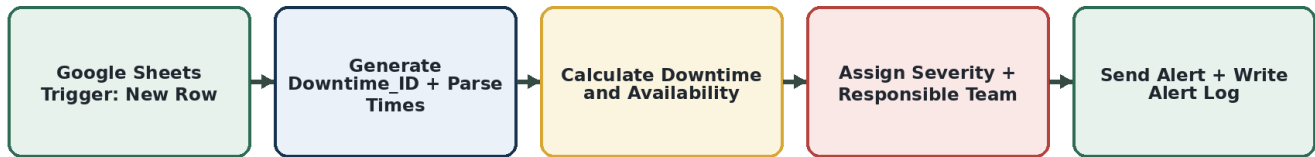
Condition	Result
Production Achievement $\geq$ 95%	Normal.
Production Achievement 90% to $<$ 95%	Warning / Medium.
Production Achievement 80% to $<$ 90%	High.
Production Achievement $<$ 80%	Critical or Very Critical.
Required production field missing or invalid	Data-quality exception; hold record for review.
Packing Actual significantly below Cement Actual	Packing or warehouse investigation.

Threshold disclaimer

The percentages above are prototype assumptions for demonstration. Final thresholds should be approved by DCP production, process and management owners before operational deployment.

8. n8n Workflow B - Downtime, Availability and Immediate Alerts

Workflow B - Downtime Incident and Availability



Node-by-node design

Step	n8n node	Configuration / output
1	Google Sheets Trigger	Watch Form_Downtime_Raw for Row Added.
2	Edit Fields / Set	Standardize plant, line, equipment, category and responsible-area values.
3	Code or expressions	Generate Downtime_ID and Alert_ID. Combine date and time fields into timestamps.
4	If - Ongoing?	If the incident is ongoing, send an immediate provisional alert. If restored, calculate the final duration.
5	Calculate	Downtime Hours = End Timestamp - Start Timestamp; Actual Runtime = Planned Runtime - Downtime; Availability = Actual Runtime / Planned Runtime x 100.
6	Validation	Check downtime does not exceed planned runtime and all mandatory reason/ownership fields are present.
7	Severity logic	Combine downtime duration, availability, estimated output loss and safety risk.
8	Switch	Route to the responsible stakeholder; critical events also route to Plant Manager.
9	Google Sheets - Append Row	Append clean incident to Downtime and validation result to Data_Quality_Validation.
10	Email / Teams + Alert Log	Send the alert and write the notification record.

Availability calculations

KPI	Calculation
Downtime Hours	Downtime_End_Timestamp - Downtime_Start_Timestamp
Actual Runtime Hours	Planned_Runtime_Hours - Downtime_Hours
Availability (%)	Actual_Runtime_Hours / Planned_Runtime_Hours x 100

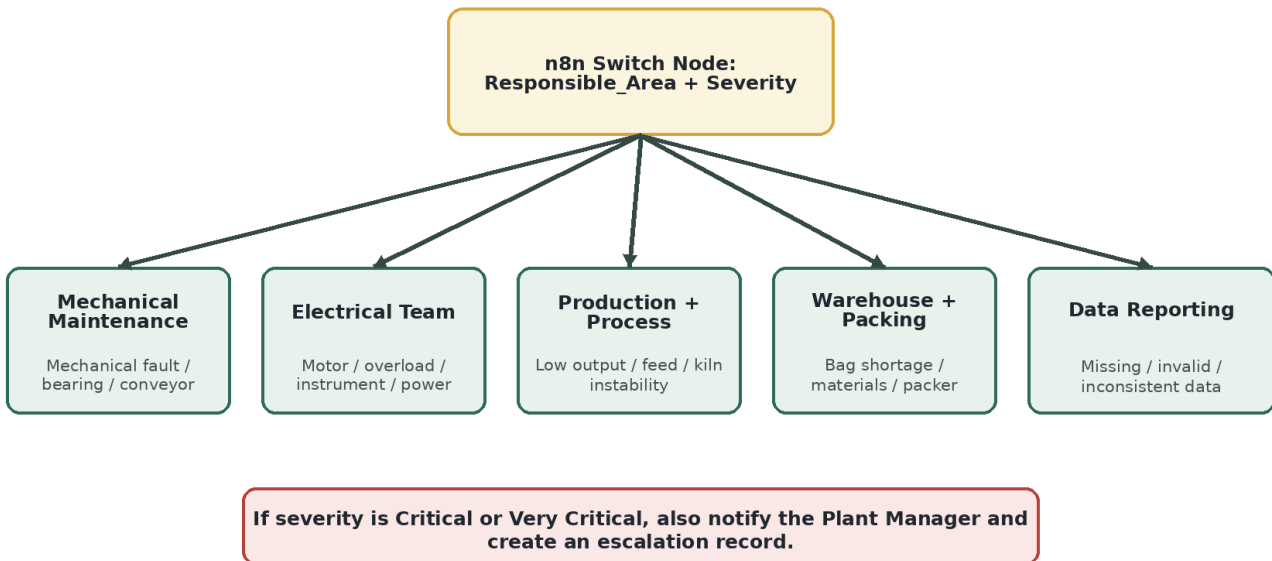
Stakeholder alert structure

Field	Example
Plant / Line	Obajana - Line 2

Field	Example
KPI	Availability
Acceptable Threshold	85%
Actual Result	78%
Planned Runtime	24.00 hours
Actual Runtime	18.72 hours
Downtime	5.28 hours
Severity	Very Critical
Possible Cause	Mechanical bearing failure
Responsible Area	Mechanical Maintenance
Recommended Action	Inspect bearing assembly, raise work order and confirm restoration.
Alert Status	Open

9. n8n Exception Routing and Escalation

Exception Routing Logic



Routing matrix

Responsible area / cause	Primary recipient	Additional escalation
Mechanical fault, bearing, belt, conveyor	Mechanical Maintenance Team	Plant Manager when severity is Critical or Very Critical.
Motor, overload, electrical trip, instrumentation	Electrical Team	Plant Manager for critical line impact.
Kiln instability, feed interruption, process deviation	Production and Process Engineering	Plant Manager when output or availability threshold is critical.
Bag shortage, material shortage, packing bottleneck	Warehouse and Packing / Logistics	Production Manager; Plant Manager for prolonged loss.
Power supply, generator or utility failure	Utilities Team	Electrical and Plant Manager as required.
Missing, invalid or inconsistent data	Data Reporting / KPI Owner	Hold publication; escalate unresolved failure.
Safety risk	HSE / Safety Team	Immediate Plant Manager escalation regardless of output effect.

Alert duplication control

- Use Alert_ID as the unique key in Alert_Log.
- Before sending an alert, look up Alert_ID or the open incident key to confirm that the same alert has not already been sent.
- For ongoing incidents, send status updates only when severity changes, the elapsed time crosses a configured threshold, or the update interval is due.
- Record recipient, channel, timestamp, delivery status and acknowledgement status.

10. n8n Workflow C - Exception Update, Resolution and Closure

Workflow C - Exception Update and Closure



Node-by-node design

Step	n8n node	Configuration / output
1	Google Sheets Trigger	Watch Form_Exception_Closure_Raw for Row Added.
2	Google Sheets - Lookup	Search Alert_Log using Alert_ID.
3	If - Alert found?	Reject or flag unknown Alert IDs for manual review.
4	Switch - Update Type	Branch to Acknowledgement, Progress Update, Resolution or Escalation.
5	Update Row	Update Alert_Log and the related Downtime or Daily_Production record.
6	Recalculate	For restored downtime, calculate final downtime and final availability.
7	Validation	Confirm required root-cause and corrective-action fields are complete before closure.
8	If - Management approval	Hold closure until approval where required.
9	Email / Teams	Send confirmation to the responder, supervisor and Plant Manager for critical exceptions.
10	Audit record	Store response time, closure time and final status.

Prefilled closure link

When an alert is sent, n8n should include a prefilled Google Form link containing Alert_ID, Plant_ID, Production_Line, Exception_Type and Responsible_Team. This reduces typing errors and ensures that the closure response maps to the correct record.

11. Daily, Weekly, Monthly and On-Demand Reporting Workflows

Daily operational report

- Use the Schedule Trigger to run after the daily reporting deadline. The Schedule Trigger supports configured daily, weekly and monthly intervals; the n8n workflow timezone must be set correctly. [6]
- Retrieve all Daily_Production, Downtime and validation records for the reporting date.
- Calculate plant and line totals, open exceptions and unresolved critical alerts.
- Send a compact Plant Manager highlight report.

Weekly performance review

- Use seven-day operating periods because cement plants normally operate continuously.
- Calculate weekly Production Achievement from total actual output divided by total target, not a simple average of daily percentages.
- Calculate weekly Availability from total actual runtime divided by total planned runtime.
- Highlight repeat downtime causes, exception counts, overdue actions and the weakest plant/line.

Monthly management review

- Compare plants and lines across production, availability, downtime, data quality and exception closure.
- Identify recurring root causes and response-time trends.
- Separate operational underperformance from data-quality failures.

Optional on-demand report

- Use an n8n Form Trigger, Webhook or chat interface to receive a requested plant and date range.
- Read matching rows from the clean sheets, calculate the KPI summary and return the formatted report.
- This is an additional historical reporting feature. It is not the primary exception-monitoring workflow.

12. Critical Field Mapping Between Forms and Clean Tables

Daily Production mapping

Google Form field	Clean column	Transformation
Timestamp	Submission_Timestamp	Retain original Form timestamp.
Reporting Date	Date	Convert to ISO date.
Plant ID	Plant_ID	Validate against Plant_Master.
Production Line	Production_Line	Validate against Plant_Master / line reference.
Clinker Target	Clinker_Target_Tons	Numeric.
Clinker Actual Output	Clinker_Actual_Tons	Numeric.
Cement Target	Cement_Target_Tons	Numeric.
Cement Actual Output	Cement_Actual_Tons	Numeric.
Packing Actual Output	Packing_Actual_Tons	Numeric.
Known Production Constraint	Reported_Constraint	Map to responsible-area rule where applicable.
Submitted By / Email	Submitted_By / Submitter_Email	Retain for accountability.

Downtime mapping

Google Form field	Clean column	Transformation
Incident Date + Start Time	Downtime_Start_Timestamp	Combine into datetime.
End Date + End Time	Downtime_End_Timestamp	Combine when restored.
Plant ID	Plant_ID	Validate against Plant_Master.
Production Line	Production_Line	Validate against line reference.
Equipment ID	Equipment_ID	Standardize uppercase.
Planned Runtime	Planned_Runtime_Hours	Numeric, zero to twenty-four.
Downtime Type	Downtime_Type	Planned / Unplanned.
Downtime Reason	Downtime_Reason	Controlled list.
Responsible Area	Responsible_Area	Controlled list used by Switch node.
Estimated Output Loss	Estimated_Output_Loss_Tons	Numeric.
Immediate Action Taken	Immediate_Action	Text.
Safety Risk	Safety_Risk_Flag	Yes / No.

Closure mapping

Google Form field	Target table / column	Action
Alert ID	Alert_Log.Alert_ID	Lookup key.
Current Status	Alert_Log.Alert_Status	Update.
Confirmed Root Cause	Alert_Log.Confirmed_Root_Cause	Update.
Corrective Action Taken	Alert_Log.Corrective_Action	Update.
Restoration Date + Time	Downtime.Downtime_End_Timestamp	Update and recalculate.
Corrected KPI Value	Relevant clean table	Update only after validation and approval.
Closure Recommendation	Alert_Log.Alert_Closure_Status	Close, monitor, escalate or reopen.

13. Implementation and Testing Checklist

Build

- Create the three Forms and one response workbook.
- Create the clean tables, reference tables, qualitative rule table and Alert_Log.
- Use identical column naming across Google Sheets, Excel and Power BI wherever possible.
- Create Google Sheets credentials in n8n and configure the three Row Added triggers.
- Build Workflow A, Workflow B and Workflow C separately before adding scheduled reports.
- Create stakeholder email or Teams templates using the highlight format.

Test cases

Test	Expected result
Valid daily production submission	Clean row appended; KPIs calculated; no alert when thresholds pass.
Production achievement below threshold	Production exception created and routed.
Missing required numeric value	Validation failure; record held or flagged; Data Reporting notified.
Mechanical downtime incident	Downtime and availability calculated; Mechanical Team notified.
Critical safety risk	Immediate escalation to Safety and Plant Manager.
Duplicate Form submission	Duplicate detected using plant, line, date and submission status.
Closure response with valid Alert ID	Alert and linked incident updated.
Unknown Alert ID	Manual review exception created; no unrelated record updated.
Critical alert not acknowledged	Reminder and escalation sent after the configured response window.
Daily scheduled run	Plant Manager summary contains only the selected date and current open alerts.

Minimum controls

- Never overwrite the raw Form response.
- Use unique IDs and lookup checks before every update.
- Protect static reference sheets and calculated columns.
- Restrict closure and corrected-value access to authorized users.
- Log every alert, update, correction and closure timestamp.
- Back up or export the workbook regularly.

14. Proof-of-Concept Scope and Future DCP Integration

This design is appropriate for a competition proof of concept using simulated operational data and human submissions. Google Forms provides structured daily and incident data without requiring access to DCP internal systems.

Prototype capability

- Near-real-time notification after a Form response reaches Google Sheets and triggers n8n.
- Consistent KPI calculations and validation rules.
- Stakeholder routing, escalation and closure audit trail.
- Daily, weekly, monthly and on-demand management reporting.

Future industrial integration

- Replace manual Forms for machine-generated data with direct SCADA, PLC, historian, ERP, laboratory and maintenance-system integrations.
- Retain Forms for human explanations, exceptional manual events and corrective-action confirmation.
- Move the operational data store from spreadsheets to a governed database for scale, concurrency and reliability.
- Obtain official KPI thresholds, equipment hierarchies, stakeholder contacts and escalation service levels from DCP owners.

Important distinction

The prototype can automate alerts quickly after structured Form submissions, but it should be described as near-real-time operational reporting. True plant real-time monitoring requires direct machine and enterprise-system integration.

15. Official References

[1] Google Docs Editors Help - Choose where to save form responses

<https://support.google.com/docs/answer/2917686?hl=en>

[2] n8n Docs - Google Sheets Trigger

<https://docs.n8n.io/integrations/builtin/trigger-nodes/n8n-nodes-base.google-sheet-trigger>

[3] n8n Docs - Google Sheets node

<https://docs.n8n.io/integrations/builtin/app-nodes/n8n-nodes-base.google-sheets>

[4] n8n Docs - If node

<https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.if>

[5] n8n Docs - Switch node

<https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.switch>

[6] n8n Docs - Schedule Trigger

<https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.schedule-trigger>

[7] n8n Docs - Microsoft Excel node

<https://docs.n8n.io/integrations/builtin/app-nodes/n8n-nodes-base.microsoft-excel>

[8] n8n Docs - Microsoft Teams node

<https://docs.n8n.io/integrations/builtin/app-nodes/n8n-nodes-base.microsoft-teams>

End of Blueprint